

Markov chain Monte Carlo

Build a random walk whose long-run visits follow the posterior, run it, and average over the path.

THE QUESTION

Rejection and importance sampling need a proposal close to p . How do we sample a posterior known only up to its constant, in many dimensions?

THE INTUITION

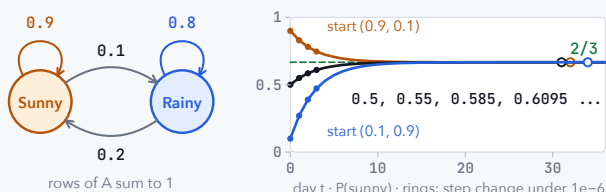
Propose a nearby point and accept it with a probability set by $\tilde{p}(x')/\tilde{p}(x)$, so Z cancels. The walk settles into p ; its draws are correlated but still average correctly.

KEY TAKEAWAY

Accept with $\min\left(1, \frac{\tilde{p}(x')q(x|x')}{\tilde{p}(x)q(x'|x)}\right)$. Then drop burn-in, run several chains and count effective samples.

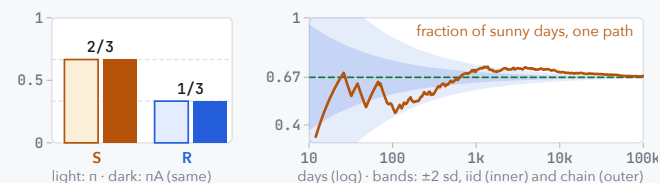
unnormalised $\tilde{p}(\theta)$ → propose $\theta' \sim q$ → accept or stay → stationary law p → burn-in, ESS → average f

1 A Markov chain forgets where it started



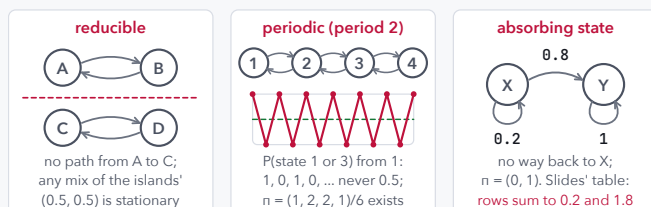
$P(X_{t+1} = j | X_t = i) = A_{ij}$, the same A at every step. The state distribution is a row vector: $\pi_{t+1} = \pi_t A$, so $\pi_t = \pi_0 A^t$. Both rows of A^{100} are $(0.6667, 0.3333)$, and the gap to $2/3$ shrinks by the second eigenvalue, 0.7 , each day.

2 The stationary distribution solves $\pi = \pi A$



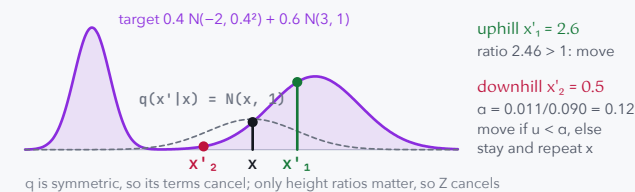
π is the left eigenvector of A for eigenvalue 1: $\pi_1 = 0.9\pi_1 + 0.2\pi_2$ and $\pi_1 + \pi_2 = 1$ give $(2/3, 1/3)$. Ergodic theorem: path averages $\frac{1}{T} \sum_t f(X_t) \rightarrow \mathbb{E}_\pi[f]$. One path: 69.6% sunny after 10,000 days, 67.4% after 100,000. Days are correlated (lag 1: 0.70), so 10,000 days count as $10^4 \cdot 0.3/1.7 \approx 1765$ independent ones.

3 When a chain does not settle



Irreducible: every state reachable from every other. Aperiodic: return times have gcd 1 (one self-loop is enough). On a finite chain these give a unique π and $\pi_t \rightarrow \pi$ from any start. The slides' reducible table has $P(X|Y) = 0.8$ and $P(Y|Y) = 1$; its arrow $X \rightarrow Y$ fixes it. A Metropolis chain whose q can reach everywhere $p > 0$ is irreducible, and its chance of staying put makes it aperiodic.

4 Metropolis-Hastings: accept, or stay put

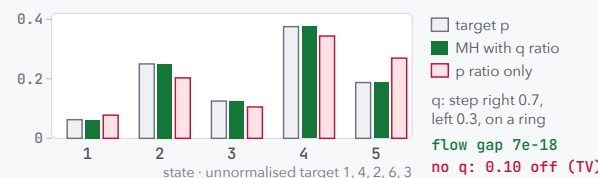


Target $p = \tilde{p}/Z$. From x draw $x' \sim q(x' | x)$ and accept with

$$\alpha(x, x') = \min\left(1, \frac{\tilde{p}(x')q(x|x')}{\tilde{p}(x)q(x'|x)}\right),$$

else repeat x . A symmetric q , such as $x' = x + \varepsilon$, cancels: $\alpha = \min(1, \tilde{p}(x')/\tilde{p}(x))$. The slides state that condition with p ; it is q .

5 Detailed balance makes p stationary



$T(x \rightarrow x') = q(x'|x)\alpha(x, x')$ gives $p(x)T(x \rightarrow x') = \min(p(x)q(x'|x), p(x')q(x|x')) = p(x')T(x' \rightarrow x)$; sum over x and $pT = p$. On five states $\pi T = \pi$ exactly and 200,000 steps land within 0.003. Drop the q ratio and the lopsided proposal wins.

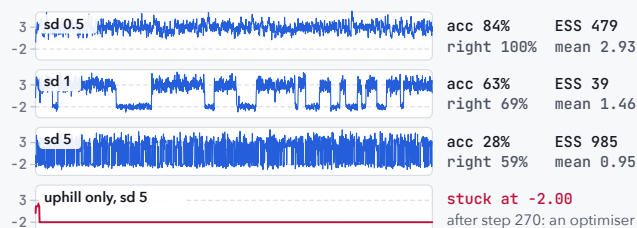
6 Metropolis-Hastings in NumPy

```
def mh(log_pt, x, n, step, rng):
    lp, out, acc = log_pt(x), [], 0
    for _ in range(n):
        z = rng.normal(size=x.shape)
        xn = x + step * z # symmetric q
        lq = log_pt(xn)
        # other q: add log q(x|xn)
        # - log q(xn|x) to lq - lp
        if np.log(rng.random()) < lq - lp:
            x, lp = xn, lq
            acc += 1
        out.append(x) # repeats count
    return np.array(out), acc / n
S, rate = mh(log_pt, np.zeros(2), 20000,
             0.2, np.random.default_rng(0))
S = S[500:] # drop burn-in (panel 10's run)
```

Check the chain, then trust its averages

Poor proposals, a transient start and correlated draws are the usual failures. Diagnostics catch them; gradients (HMC) avoid many.

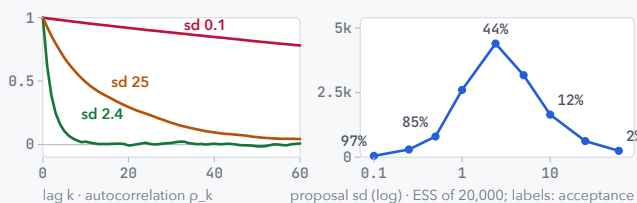
7 Proposal width trades acceptance against mixing



10,000 steps from 0.1 (uphill: 1000) · right: share above 0, target 60% · target mean 1.00

The notebook's two-peak target from 0.1. At sd 0.5 most steps are accepted but the walk stays on the right peak; sd 5 accepts 28%, crosses 894 times and gets the 60 : 40 split. Uphill-only moves (the notebook's first sampler) are hill climbing: they stop at a mode. Aim near 44% in 1D, 23% in many dimensions (Roberts et al. 1997).

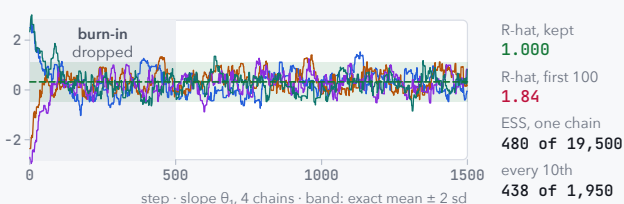
8 Correlated draws: count effective samples



$$ESS = \frac{N}{1 + 2 \sum_{k \geq 1} \rho_k}, \quad MC \text{ error} = sd / \sqrt{ESS}$$

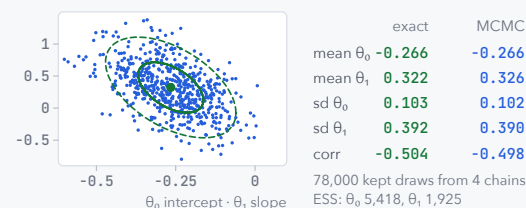
ρ_k is the lag- k autocorrelation; the sum stops when pairs $\rho_{2m} + \rho_{2m+1}$ turn negative. On $\mathcal{N}(0, 1)$, sd 0.1 accepts 97% but gives ESS 52 and a mean of -0.22 .

9 Burn-in, several chains, thinning



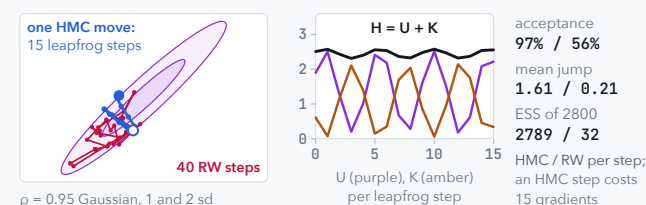
Four chains from far corners reach the ± 2 sd band of the panel 10 posterior within 33 to 69 steps; drop a burn-in (the slides discard the first $N_1 - 1$). $\hat{R}$, the square root of pooled over within-chain variance, compares the chains; above 1.01, keep running. Keeping every 10th draw saves memory, keeps 91 to 96% of the ESS and never adds information.

10 MCMC recovers the exact regression posterior



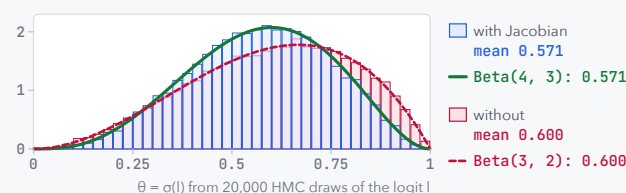
The Bayesian linear regression sheet's five points, prior $\mathcal{N}(0, I/2)$, noise sd 0.2: the exact posterior is $\mathcal{N}(m_N, S_N)$. Metropolis sees only $\log \tilde{p}(\theta) = -\|y - X\theta\|^2 / (2\sigma^2) - \theta^T \theta$. The slope mean is off by 0.004, inside its MC error of 0.009.

11 HMC: far proposals, still accepted



Momentum $r \sim \mathcal{N}(0, I)$, $H = U(\theta) + r^T r / 2$, $U = -\log \tilde{p}$. Leapfrog L times: $r \leftarrow r - \frac{\epsilon}{2} \nabla U$, $\theta \leftarrow \theta + \epsilon r$, $r \leftarrow r - \frac{\epsilon}{2} \nabla U$; accept with $\min(1, e^{H_{\text{old}} - H_{\text{new}}})$. Leapfrog is reversible and keeps volume (no q ratio) and nearly keeps H (drift 0.27 here). The course's hamiltorch runs accept 0.95 (a line fit) and 0.96 (a 51-weight classifier), quoted.

12 Constrained parameters: sample the logit, add the Jacobian



Coin: 3 heads, 2 tails, Beta(1, 1) prior, so Beta(4, 3). HMC on θ leaves (0, 1) (34% of trajectories from 0.5). With $\theta = \sigma(l)$:

$$\log p(l) = \log \tilde{p}(\sigma(l)) + \log \sigma(l) + \log(1 - \sigma(l)).$$

The notebook drops the last two terms: Beta(3, 2). Predict by averaging probabilities over draws, not logits.

stationary law $p \rightarrow \text{accept } \min(1, \tilde{p}' q / (\tilde{p} q')) \rightarrow \text{drop burn-in, check } \hat{R} \text{ and ESS} \rightarrow \text{average } f(\theta_t) \cdot \text{HMC: far moves, high acceptance}$

Source: lecture slides "Sampling methods" (Markov chains and MCMC) and the notebooks on Markov chains, MCMC as optimisation and HMC (hamiltorch outputs quoted), nipunbatra.github.io/pml-teaching other numbers from NumPy with fixed seeds. Corrections: symmetric MH is a condition on q ; the reducible table's rows sum to 0.2 and 1.8; "stationary" is no condition for a unique π ; the notebook's other starts also reach (0.6667, 0.3333), not its printed 0.6507, 0.6733, 0.6761; "no cycles" is not aperiodicity; its Metropolis cells replot the greedy samples; the logit coin lacks the Jacobian; the classifier averages logits.